

Week 1

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1 Tannaka duality

Theorem 1.1 (Stefanich [Ste23]). The functor

$$X \longmapsto \mathrm{QCoh}(X)^\heartsuit$$

from qcqs Artin stacks with affine diagonal to presentable abelian ∞ -categories is fully faithful.

All of the abelian categories that arise in this way have compact unit. However, some examples from arithmetic geometry do not satisfy this property.

Example 1.2 (Almost mathematics). Let (A, m) be a local ring with flat idempotent maximal ideal m , i.e. a flat maximal ideal m such that $m^2 = m$. One can consider the full subcategory defined by the exact/localization sequence

$$\mathrm{aMod}_A \longrightarrow \mathrm{Mod}_A \longrightarrow \mathrm{Mod}_{A/m}.$$

The inclusion preserves colimits and compact objects. Thus the compact objects in aMod_A are precisely the finitely presented A -modules. But by Nakayama's lemma, they are all trivial. This suggests that one cannot present this category using only the geometric objects appearing in Theorem 1.1.

Therefore, the natural question posed here is how to build a useful geometric framework that captures these examples.

2 $\mathcal{C}$ -Algebraic Geometry

The guiding idea, due to Deligne [Del90], is that for any presentable abelian ∞ -category $\mathcal{C}$, one can try to develop “ $\mathcal{C}$ -algebraic geometry” by the following steps:

- (i) define affine objects as $\mathrm{CAlg}(\mathcal{C})^{\mathrm{op}}$;
- (ii) put a Grothendieck topology on $\mathrm{CAlg}(\mathcal{C})^{\mathrm{op}}$;
- (iii) take sheaves.

Classically, given a scheme S , one has an equivalence of categories

$$\left\{ \begin{array}{c} X \rightarrow S \\ \text{affine} \end{array} \right\} \simeq \mathrm{CAlg}(\mathrm{QCoh}(S)^\heartsuit)^{\mathrm{op}}.$$

Thus the affine geometry over S can be read from $\mathrm{CAlg}(\mathrm{QCoh}(S)^\heartsuit)^{\mathrm{op}}$.

The above construction is useful in several contexts. Applied to almost modules, it recovers almost mathematics; applied to functional analytic rings, it gives theories of analytic geometry.

There is a small problem: localization to open subsets is not flat in this generality. Thus one is led to consider the ∞ -category

$$\mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L).$$

In fact there is a functor

$$\mathrm{CAlg}(D(\mathbb{S})) \longrightarrow \mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L)$$

coming from the underlying presentable stable category. The question then becomes: what is the geometry on $\mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L)$? Such a geometry is not only technically hard to control; it also fails to capture some geometric examples, as the following section illustrates.

3 A computation of $\mathrm{QCoh}(B^n \mathbb{G}_m)$

The computation below follows Gaitsgory–Rozenblyum [GR17]. Fix a field k of characteristic 0, and let G be a smooth affine group. Let $B^n G$ denote the higher Artin stack, and let

$$p_n : * \longrightarrow B^n G, \quad f_n : B^n G \longrightarrow *$$

be the corresponding smooth atlas and structure morphism for $B^n G$.

3.1 The case $n = 1$

For $n = 1$, this is classical:

$$\mathrm{QCoh}(B\mathbb{G}_m) \simeq \mathcal{D}(\mathrm{Rep}(\mathbb{G}_m)) \simeq \mathrm{Fun}(\mathbb{Z}^{\mathrm{disc}}, D(k)).$$

In general, for any smooth affine group G , we have

$$\mathrm{QCoh}(BG) \simeq \mathcal{D}(\widehat{\mathrm{Rep}(G)}).$$

Here the hat denotes left completion.

3.2 The case $n \geq 2$

A first thought is to use the cover $p : * \rightarrow B^n G$. Since QCoh satisfies descent, one has

$$\mathrm{QCoh}(B^n G) \simeq \mathrm{Tot}(\mathrm{QCoh}((B^{n-1} G)^{\times \bullet})).$$

Now we want to show that

$$p^* : \mathrm{QCoh}(B^n G) \longrightarrow \mathrm{QCoh}(*)$$

is comonadic.

Proposition 3.1. [GR17, Proposition 2.3.2]

Given a Cartesian square of Artin stacks

$$\begin{array}{ccc} Y'_1 & \xrightarrow{g'} & Y_1 \\ f' \downarrow & & \downarrow f \\ Y'_2 & \xrightarrow{g} & Y_2 \end{array}$$

where g is flat and f is qcqs, the square

$$\begin{array}{ccc} \mathrm{QCoh}(Y_1)_{\leq 0} & \xrightarrow{g^*} & \mathrm{QCoh}(Y'_1)_{\leq 0} \\ f_* \downarrow & & \downarrow f'_* \\ \mathrm{QCoh}(Y_2)_{\leq 0} & \xrightarrow{g^*} & \mathrm{QCoh}(Y'_2)_{\leq 0} \end{array}$$

is commutative.

Sketch of proof. We only show that f_* sends coconnective objects to coconnective objects and that f_* preserves filtered colimits. The proof is by induction.

Indeed, for $\mathcal{F} \in \mathrm{QCoh}(Y_1)_{\leq 0}$, one has

$$f_*(\mathcal{F}) \simeq \mathrm{Tot}(f_*^i(\mathcal{F}|_{Z^i/Y_1})),$$

where $Z^\bullet \rightarrow Y_1$ is a smooth atlas and

$$f_*^i : Z^i/Y_1 \longrightarrow Y_1 \xrightarrow{f} Y_2.$$

By the induction hypothesis, each f_*^i preserves filtered colimits on coconnective objects. Since $\mathrm{QCoh}(Y_2)$ is right t -complete, this reduces the problem to proving that $\tau_{\geq m} \mathrm{Tot}$ preserves filtered colimits.

But this is also a limit computed in $\mathrm{QCoh}(Y_2)_{[m,0]}$, which is a $(1-m, 1)$ -category. In such a category, $\lim_{\Delta} \cong \lim_{\Delta \leq 1-m}$ commutes with filtered colimits. Hence Tot preserves filtered colimits as desired. $\square$

By [Lur17, Theorem 4.7.5.2], this gives the desired comonadicity (for co-connective parts). The condition in that theorem follows by applying the proposition above to the Cartesian square

$$\begin{array}{ccc} B^l G^{\times j} & \longrightarrow & B^l G^{\times i} \\ \downarrow & \lrcorner & \downarrow \\ B^l G^{\times j+1} & \longrightarrow & B^l G^{\times i+1} \end{array}$$

Thus

$$\mathrm{QCoh}(B^n G)_{\leq 0} \simeq \mathrm{Comod}_{p_* p^*}(\mathrm{Mod}_{k, \leq 0}).$$

The triviality of $\mathrm{QCoh}(B^{n+1} \mathbb{G}_m)$ in the case $n \geq 1$ follows from

$$R\Gamma(B^n \mathbb{G}_m; \mathcal{O}) \simeq k$$

because

$$p_{n+1,*} p_{n+1}^*(-) = f_{n,*} f_n^*(-) = f_{n,*} \mathcal{O}_{B^n \mathbb{G}_m} \otimes_k - = R\Gamma(B^n \mathbb{G}_m; \mathcal{O}) \otimes_k -.$$

Here the projection formula holds via a similar proof as in proposition 3.1 and the fact that the Tot-tower $\mathrm{Tot}^{\leq *} R\Gamma(B^n \mathbb{G}_m^{\times \bullet}; \mathcal{O})$ is (pro-)constant.

Remark 3.2. It should not be very hard to generalize the argument to higher quasicoherent sheaves. The general heuristic is that $(n-1)\mathrm{QCoh}(B^n \mathbb{G}_m)$ is too small to capture the full geometry. Thus, if one wants to include all of these geometric objects, one should consider ∞ -categories at all categorical levels, together with higher sheaf theory formulated in that setting.

4 Higher Presentable Categories

For a usual scheme X , one can write

$$D(X) \simeq \lim_{\mathrm{Spec} A \rightarrow X} D(A).$$

This suggests defining

$$\mathrm{Pr}_X^L := \lim_{\mathrm{Spec} A \rightarrow X} \mathrm{Pr}_A^L, \quad \mathrm{Pr}_A^L := \mathrm{Mod}_{D(A)}(\mathrm{Pr}_{\mathrm{St}}^L).$$

The question is whether higher presentable categories are the right language in which to formulate this higher sheaf theory.

The naive approach leads to an infinite hierarchy of universes. Stefanich's fix [Ste20] is to work with compact objects such as

$$\mathrm{Mod}_M(\widehat{\mathrm{Cat}}_{\mathrm{comp}})_{\kappa_0}, \quad \mathrm{Mod}_{\mathrm{Mod}_M(\widehat{\mathrm{Cat}}_{\mathrm{comp}})_{\kappa_0}}(\widehat{\mathrm{Cat}}_{\mathrm{comp}})_{\kappa_0}, \quad \dots$$

Aoki [Aok25] gives a more conceptual interpretation:

$$\mathrm{Pr}_k^L \in \mathrm{Pr}_k^L, \quad n\mathrm{Pr}_k := \mathrm{Mod}_{(n-1)\mathrm{Pr}_k}(\mathrm{Pr}_k^L),$$

and then we have

$$n\mathrm{Pr}^{\mathrm{Stefanich}} = \mathrm{colim}_{k \text{ regular}} n\mathrm{Pr}_k.$$

Thus, to simplify the theory, one may take $\kappa = \aleph_1$ and set $n\mathrm{Pr} := n\mathrm{Pr}_{\aleph_1}$.

5 Stefanich's Ring

Now we can define Stefanich's ring. There is a sequence of inclusions in $\mathbf{Pr}^L$:

$$\mathrm{CAlg}(\mathbf{Pr}) \longrightarrow \mathrm{CAlg}(2\mathbf{Pr}) \longrightarrow \mathrm{CAlg}(3\mathbf{Pr}) \longrightarrow \cdots.$$

Define

$$\mathrm{StRing} := \mathrm{colim}_n \mathrm{CAlg}(n\mathbf{Pr}) \quad \text{in } \mathbf{Pr}^L.$$

Equivalently,

$$\mathrm{StRing} \simeq \lim_{\mathrm{End}_-(1)} \mathrm{CAlg}(n\mathbf{Pr}) \quad \text{in } \mathbf{Cat}.$$

Fact 5.1. Higher QCoh gives a functor

$$\mathrm{AnStack} \longrightarrow \mathrm{StRing}^{\mathrm{op}},$$

which is conservative on $\mathbb{N}_1$ -compact objects.

Question 5.2. Does this recover the desired geometry?

The answer seems to be yes for the intended examples, but the resulting geometry is still even more difficult to control.

Theorem 5.3 (Scholze–Stefanich). For all practical purposes,

$$\mathrm{Gest} := \mathrm{StRing}^{\mathrm{op}}$$

is an ∞ -topos.

References

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